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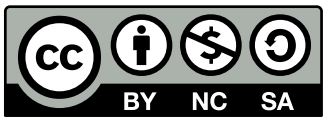
Set Propagation Techniques for Reachability Analysis

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Bachelor's Thesis
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Abstract

Reachability analysis is a fundamental problem in the analysis of dynamical systems. It involves computing the reachable set of states for a system. Starting from an initial set of states, the reachable set of a system is computed by considering the system dynamics and admissible control inputs. The reachable set contains all possible trajectories of the system. Reachability analysis is a valuable tool used to ensure that safety-critical systems function correctly in all possible scenarios. This literature-review focuses on set propagation techniques for reachability analysis. Set propagation techniques use set representations to approximate the reachable set of a system. Thus, making them a scalable approach to reachability analysis. This thesis provides background information on the most popular set representations and compares the performance of cutting-edge set propagation techniques with Matlab-based reachability toolbox CORA using concrete examples. The case studies prove that certain methods are more scalable to large-scale systems. However, their accuracy is inferior compared to other approaches.

Keywords Set Propagation, Reachability Analysis, Linear Systems, Nonlinear Systems

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Tiivistelmä

Saavutettavuusanalyysin päämääränä on selvittää tilat, joihin systeemi on mahdollista ohjata ohjaussekvenssillä jostakin tietystä alkutilasta. Saavutettavuusanalyysillä on useita käyttökohteita säätöjärjestelmissä ja robotiikassa. Yksi tärkeimmistä käyttökohteista on muodollinen verifiointi. Sen avulla pystytään varmistamaan, että systeemi toimii odotetusti ja turvallisesti kaikissa mahdollisissa tilanteissa. Tämä on hyödyllistä monissa säätötekniikan sovelluksissa, joissa systeemin tulee saavuttaa tietty tila samalla välttäen vaaralliset tilat. Vaarallisia tiloja voi olla esimerkiksi esteet autonomiselle ajoneuvolle tai robottikäsivarrelle. Saavutettavuusanalyysin suorittamiseen on olemassa monia eri menetelmiä. Tämä opinnäytetyö keskittyy joukko propagaatio tekniikkoihin.

Joukko propagaatio tekniikat ovat viime aikoina nousseet suosioon niin akateemisessa maailmassa kuin teollisuudessa. Syy tähän on niiden hyvä skaalautuvuus laajempiin systeemeihin. Tämä johtuu siitä, että nämä tekniikat käyttävät saavutettavan joukon laskemiseen geometrisia muotoja eivätkä tarkkoja numeerisia arvoja. Täten joukko propagaatio tekniikat eivät vaadi differentiaaliyhtälöiden numeerista ratkaisemista, mikä vähentää tarvittavaa laskentatehoa. Kaikkien joukko propagaatio menetelmien perusidea on samankaltainen. Alkaen valitusta alkutilasta joukko propagaatio menetelmät määrittävät systeemin seuraavat mahdolliset tilat ottaen huomioon systeemin dynamiikan, mahdolliset ohjaukset sekä mahdolliset. Lopputulos riippuu muun muassa käytetystä algoritmista, alkutilasta, joukkoesityksen geometrisestä muodosta sekä parametreistä kuten askelkoosta.

Tämä kirjallisuuskatsaus esittelee käytetyimpiä joukkojen esitystapoja sekä joukko propagaatio tekniikan perusteet. Työssä käsitellään aiheeseen liittyviä tieteellisiä julkaisuja ja niiden saavutuksia. Työ keskittyy tarkastelemaan erilaisia joukko propagaatio menetelmiä sekä lineaarisille että epälineaarisille systeemeille. Näiden esiteltyjen menetelmien tehokkuutta, vahvuuksia ja heikkouksia analysoidaan eri esimerkkisysteemien avulla käyttäen Matlabin CORA-työkalua. Erityisesti tarkastellaan eri menetelmien skaalautuvuutta systeemeihin, joissa on suuri määrä tilamuuttujia.

Saadut tulokset osoittavat, että lineaarisille järjestelmille tietyt menetelmät skaalautuvat paremmin laajempiin systeemeihin. Toisaalta nämä menetelmät eivät tuota yhtä tarkkaa saavutettavaa joukkoa kuin ne menetelmät, jotka eivät skaalaudu yhtä hyvin. Epälineaarisille systeemeille tulokset ovat samankaltaisia. Toinen testatuista metodeista on nopeampi, mutta tuottaa epätarkempia tuloksia. Vastaavasti, tarkemman metodin avulla saadaan laadukkaampia tuloksia, mutta tämä metodi vaatii enemmän laskenta-aikaa. Yhteenvetona, käytettävä joukko propagaatio metodi täytyy valita punnitsemalla juuri toteutettavan applikaation tarpeita. Menetelmän valinnassa tulee ottaa huomioon tarkkuus sekä laskenta-aika.

Avainsanat saavutettavuusanalyysi, joukkopropagaatio, lineaariset systeemit, epälineaariset systeemit

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Symbols and abbreviations

Symbols

$\mathbb{R}$	Set of natural numbers
$\mathbb{N}$	Set of real numbers
$\mathbf{A}$	System matrix
$\mathbf{B}$	Input matrix
$\mathcal{U}$	Set of possible inputs
$\mathcal{X}_0$	Set of initial states
$\mathcal{H}$	Reachable set
$x(t)$	State vector
$u(t)$	Input vector

Operators

A $\oplus$ B	Minskowski sum of sets A and B
$\dot{x}$	Time derivative of x
CH(A,B)	Convex hull of sets A and B

Abbreviations

LTI	Linear time-invariant
ODE	Ordinary Differential Equation
PDE	Partial Differential Equation
HJ	Hamilton-Jacobi (Equations)

1 Introduction

Reachability analysis is a key tool in the study of dynamical systems, especially in robotics and control problems. It involves computing the set of all states that a dynamical system can reach over time with possible control sequences, given initial set of states and system dynamics. The current approach to analyzing dynamical systems in industry is mainly based on computing possible trajectories for dynamical systems by simulating the system under different circumstances [1]. However, this method does not guarantee the safety of the system as it can only investigate a finite number of scenarios. Thus, an unsafe trajectory may not be discovered. To exhaustively explore all possible states that the system might reach, we need to rely on reachability analysis. Reachability analysis is a crucial part in ensuring the safe behavior of the system in safety-critical applications, such as autonomous vehicles, airplane guidance systems, robotics, and medical devices [2].

Since it is not possible to compute the exact reachable set for almost any system, the reachable sets are computed as over-approximations that are guaranteed to contain the exact reachable set. This is useful in applications such as formal verification, state-estimation, and controller synthesis. In formal verification, we can verify the safety of the system by checking whether the over-approximated reachable set intersects with unsafe set of states. [1, 2].

Reachability analysis can be performed using many different approaches. One approach is to numerically solve Hamilton-Jacobi (HJ) equations to approximate the reachable set of a system. However, the computational expense of numerically solving HJ-equations increases rapidly with the number of state variables. Another approach to reachability analysis is barrier certificates. A barrier certificate is essentially a function that defines a barrier that encloses the reachable set of the system. Barrier certificates can be useful in proving that the system cannot reach unsafe sets that lie on the other side of the barrier. [1]. This thesis does not examine the aforementioned approaches any closer, but instead focuses on set propagation techniques for reachability analysis.

Real-world systems often have a substantial number of state variables. The number of state variables is used as a synonym with system dimension. Due to their scalability to higher-dimensional systems, set propagation techniques for reachability analysis have recently gained popularity in the industry. Computing reachable sets using set propagation is similar to numerically solving ordinary differential equations (ODEs), but instead of exact numbers, the solutions are represented with set representations. Set representations are essentially n-dimensional geometrical shapes, such as ellipsoids, polytopes or zonotopes. The dimension of the chosen set representation is naturally equal to the system dimension.

1.1 Continuous Linear Systems

The set propagation techniques for the reachability analysis of linear time-invariant (LTI)-systems have been studied extensively. One usually resorts to over-approximation methods, due to their simplicity to compute compared to under-approximations. Most of these techniques rely on convex set representations such as zonotopes [3, 4],

ellipsoids [5] and support functions [6].

Over-approximating the reachable set at each timestep throughout the iteration can lead to massive errors in the final reachable set. This is known as the wrapping effect. The wrapping effect can be mitigated by various techniques, that avoid the propagation of error. An algorithm that mitigates the wrapping effect in the computation of reachable sets for LTI-systems is introduced in [3]. The algorithm restructures the computations required to prevent over-approximation at each iteration. This algorithm is capable of computing tight over-approximations of reachable sets of 100-dimensional systems.

In real-world applications, system parameters are almost always uncertain. Therefore, the implementation of robust methods that can handle uncertainty is crucial. In [7], a reachability algorithm is proposed for linear systems with uncertain parameters. This approach uses non-convex polynomial zonotopes to tightly over-approximate the reachable set of system. The paper also extends this method to linear time-varying systems.

The computational complexity of standard reachability methods often grows exponentially with respect to the system dimension. Therefore, linear systems with roughly more than thousand state variables require more advanced methods for reachability analysis that reduce this complexity. Order reduction methods based on the Krylov subspace approximation have proven to be one of the most successful methods for reachability [8]. In [8], this method is extended to work with arbitrary inputs. A numerical verification for affine systems with even billion dimensions is achieved in [9] using the Krylov methods by exploiting the problem structure.

1.2 Nonlinear Systems

Systems with nonlinear dynamics are much trickier to analyze as useful properties such as linear mapping and superposition principle are no longer valid. Therefore, many approaches rely on local linearization of the model in order to take advantage of methods used for linear systems [10]. The linearization error considered by enlarging the reachable set at each time step.

In [11], the reachability analysis of nonlinear systems is based on abstraction to polynomial differential inclusions. Furthermore, a non-convex set representation called polynomial zonotopes is presented to capture the non-convexity of the reachable set more accurately. This makes the method considerably more accurate

One of the most recent promising approaches to reachability analysis of nonlinear systems involves Koopman operator linearization. Bak et al. propose a black box solution to the resulting reachability problem after linearization [12]. However, the Koopman invariant subspace required for this has yet to be discovered.

1.3 Thesis Structure

The purpose of this thesis is to give the reader a thorough understanding of cutting-edge set propagation techniques, comparing the performance of various methods, highlighting their advantages and limitations, and possible areas for improvement, where

relevant. This thesis mainly focuses on over-approximation techniques of forward reachable sets due to the fact that they are more commonly used and more researched. First, the most popular set representations for set-based reachability analysis are introduced, followed by an overview of the most relevant set propagation techniques. Lastly, the performance of different set propagation techniques is examined through concrete examples using existing tools for reachability analysis such as Matlab-based CORA [13].

2 Preliminaries

This section introduces the basics of reachability analysis and set representations, which are required to understand set propagation techniques. Different set representations are analyzed in the context of reachability analysis to assess their suitability and effectiveness. Section 2.1 presents the fundamental problem statement for reachability analysis of continuous systems. Sections 2.2 and 2.3 provide an overview of the most popular convex and nonconvex set representations, respectively.

2.1 Reachability Analysis

The main goal of reachability analysis is to compute the reachable set of the system. The reachable set exhaustively covers all the possible trajectories of the system from an initial set of states considering system dynamics, input, and disturbances. Figure 1 illustrates the reachable set of car model.

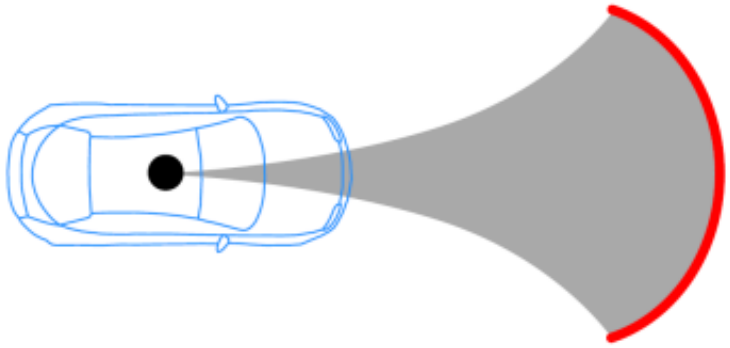


Figure 1: The reachable set for a car model representing the position of the car in x-y plane [7]. The final reachable set is marked in red and the whole reachable set in gray.

Let us define the formal problem statement for the forward reachability analysis. Given an LTI-system with dynamics defined as

$$\dot{x}(t) = \mathbf{A}\,x(t) + \mathbf{B}\,u(t) \tag{1}$$

, where $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\mathbf{B} \in \mathbb{R}^{n\times m}$, $u(t) \in \mathbb{R}^m$ is the control input and $x(t) \in \mathbb{R}^n$ is the state of the system [7]. The solution trajectory for the LTI-system at time point $t \in \mathbb{R}$, starting from an initial set of $x_0 \in \mathbb{R}^n$ with input trajectory $u(\cdot)$ is $\xi\big(t, x(0), u(\cdot)\big)$. Given an initial set of states $\mathcal{X}_0 \subset \mathbb{R}^n$ and a set of possible inputs $\mathcal{U} \subset \mathbb{R}^n$ the forward reachable set for a system at time t is defined as

$$\mathcal{R}(t) = \xi\big(t, x(0), u(\cdot)\big) \;\Big|\; x(0) \in \mathcal{X}_0,\ \forall\, s \in [0,t] : u(\theta) \in \mathcal{U} \tag{2}$$

[14]. The reachable set of states for the entire time horizon can be obtained as a union of reachable sets at time t.

$$\mathcal{R}([0,r]) = \bigcup_{t\in[0,r]} \mathcal{R}(t). \tag{3}$$

[10]. Section 4 introduces the basic framework for computing the reachable set using set propagation techniques.

2.2 Convex Set Representations

In order to compute reachable sets for a system, the sets must be represented efficiently on a computer. In practice, this usually means over-approximating the true reachable set with a set of geometric shapes, such as ellipsoids, zonotopes or polytopes. A convex set is a set in which, for any two points you choose within the set, the entire line segment connecting those two points also lies entirely within the set. The reachable sets of linear systems are often convex [1]. Therefore, convex set representations can accurately describe the dynamics of these sets.

The use of a different set representation can significantly improve the computation time and precision of set-based reachability algorithms [15]. The reason for this is that the set operations required in the computation of reachable sets are more efficient with certain set representations. The most used set operations for reachability algorithms include linear transformation, Minkowski sum [16], convex hull [17] and intersection. Minkowski sum is essentially an addition of two sets. Whereas, convex hull of two sets returns the smallest convex set that encloses both sets. This section introduces the most popular convex set representations, including ellipsoids, polytopes, zonotopes, ellipsotopes and support functions. The relevant properties for reachability algorithms are discussed for each set representation.

2.2.1 Intervals

Intervals (rectangles) are a simple set representation that is easy to implement and efficient for computation. However, intervals suffer from being overly conservative, resulting in large over-approximations. Intervals are used mainly in real-time control systems, which require more computational efficiency than precision. Other common use-cases for intervals are defining the initial set of states for a system and over-approximating some other set representation. The initial set of states encloses all the states from which the system can start from. An interval set is illustrated in Figure 2. The formal definition for a real-valued multidimensional interval is

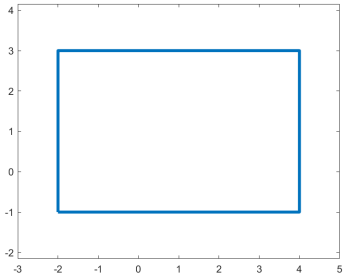


Figure 2: An example of an interval set [13]

$$I := \{x \in \mathbb{R}^n \mid \underline{x}_i \leq x_i \leq \overline{x}_i \; \forall i = 1, \ldots, n\} \tag{4}$$

[1].

2.2.2 Ellipsoids

Ellipsoidal set representations provide a compact and smooth representation for reachable sets. Figure 3 shows an ellipsoidal set representation. The downside of ellipsoids is that they are not closed under key set operations, such as Minkowski sum and intersection [15]. This means that the result of these operations is not an ellipsoid. Consequently, these operations require an approximation of the result, which adds error to the reachable set every time the operation is applied. The formal definition for ellipsoids is

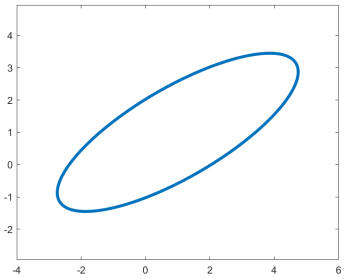


Figure 3: An example of an ellipsoidal set [13]

$$\mathcal{E} := \left\{ x \in \mathbb{R}^n \mid (x-q)^T Q^{-1} (x-q) \leq 1 \right\} \quad (5)$$

, where $q \in \mathbb{R}^n$ is the origin and $Q \in \mathbb{R}^{n \times n}$ is a positive definite symmetric matrix that defines the shape of the ellipsoid [1].

2.2.3 Polytopes

Polytopes are essentially n-dimensional polygons that offer a precise and flexible representation for linear systems with moderate dimensions. A polytope is displayed in Figure 4. The limitation for polytopes is that operations, such as intersection and Minkowski sum are expensive to compute [18]. For polytopes, there is two types of representations, H-representation, which are defined as

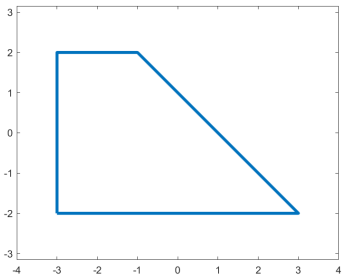


Figure 4: An example of a polytope [13]

$$P := \left\{ x \in \mathbb{R}^n \mid Ax \leq b \right\}, \; A \in \mathbb{R}^{q \times n}, \; b \in \mathbb{R}^q \quad (6)$$

and V-representation defined by convex hull of a set of vertices:$\{v_1, \ldots, v_r\} \subseteq \mathbb{R}^m$

$$P := \mathrm{CH}\left(\mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(r)}\right) \quad (7)$$

[1].

2.2.4 Zonotopes

A zonotope is a symmetric n-dimensional polytope (see Figure 5. They can be compactly represented as

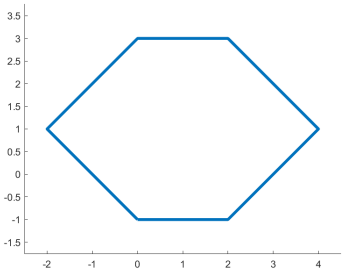


Figure 5: An example of a zonotope [13]

$$Z = c|G := \left\{ c + \sum_{i=1}^{p} \beta_i g^{(i)} \;\middle|\; \beta_i \in [-1, 1] \right\} \quad (8)$$

, where $c \in \mathbb{R}^n$ is the center of the zonotope and $G \in \mathbb{R}^{n \times p}$ is a matrix consisting of p generators

$g^{(i)} \in \mathbb{R}^n, i \in \{1, ...p\}$ [19]. The order of a zonotope is $\frac{p}{n}$. Since set operations on higher-order zonotopes are computationally more expensive, the order of zonotopes is often reduced by over-approximating a higher-order zonotope with a lower-order zonotope [20].

Unlike polytopes, zonotopes are closed under Minskowski sum and linear transformation, which are essential operations when computing reachable sets. Since zonotopes have many useful characteristics for the computation of reachable sets, they are one of the most popular set representations in reachability analysis [4].

2.2.5 Ellipsotopes

In [21], a set representation called ellipsotopes, combining the advantages of ellipsoids and zonotopes, is proposed. Ellipsotopes can efficiently model high-dimensional systems while remaining computationally tractable. Unlike ellipsoids, ellipsotopes allow efficient approximation of operation such as intersections and Minkowski sums. Given $c \in \mathbb{R}^n$, $\mathbf{G} \in \mathbb{R}^{n \times m}$, $\mathbf{A} \in \mathbb{R}^{k \times m}$, and $b \in \mathbb{R}^k$, and let $\mathcal{I}$ be a valid index, ellipsotopes are defined as:

$$\mathcal{E}_p(c, G, A, b, \mathcal{I}) = \{c + G\beta \mid \|\beta(J)\|_p \leq 1 \; \forall \, J \in \mathcal{I} \text{ and } A\beta = b\} \subset \mathbb{R}^n \tag{9}$$

[21].

2.2.6 Support Functions

Instead of representing the reachable set with a geometric shape, support functions characterize convex sets by measuring the maximum extent of a set in a finite number of directions. Support functions can be used to represent any convex set [15]. The sets represented by support functions are closed under linear transformations, Minkowski sums, and convex hulls. For these reasons, support functions are used in many reachability algorithms [6]. Given a convex set $\Omega \subseteq \mathbb{R}^d$. and a direction ℓ, the support function of the set is defined as:

$$\rho_\Omega : \mathbb{R}^d \to \mathbb{R} \quad \ell \mapsto \max_{x \in \Omega} \ell \cdot x \tag{10}$$

[22].

2.3 Non-convex Set Representations

Non-convex set representations are more obscure than convex sets and do not fulfill the definition of convexity. Non-convex sets provide more accurate approximations for the reachable set when the dynamics of the system are nonlinear [1]. This section introduces nonconvex set representations, such as Taylor models, polynomial zonotopes, star sets and level sets.

2.3.1 Taylor Models

Taylor models (see 6) use the Taylor approximation enlarged by an error interval to accurately represent a non-convex set. Intuitively, they are n-dimensional Taylor polynomials expanded by an n-dimensional interval. Taylor models are ideal for analyzing nonlinear systems, but require higher computational cost compared to simpler set representations [1]. Taylor models offer an accurate and scalable approach for the reachability analysis of nonlinear hybrid systems [23].

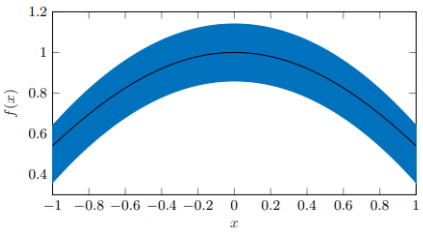


Figure 6: An example of a Taylor model [13]

2.3.2 Polynomial Zonotopes

As zonotopes fail to accurately represent the non-convexity of nonlinear dynamics, a non-convex generalization of zonotopes is proposed in [11]. Polynomial zonotopes (see Figure 7) can represent nonlinear dynamics more accurately. Similar to zonotopes, order reduction techniques can also be applied to polynomial zonotopes to improve computational efficiency. The formal definition for a polynomial zonotope is

$$\mathcal{PZ} = \left\{ c + \sum_{i=1}^{h} \left(\prod_{k=1}^{p} \alpha_k^{E_{k,i}} \right) G(\cdot, i) + \sum_{j=1}^{q} \beta_j \, GI(\cdot, j) \;\middle|\; \alpha_k, \beta_j \in [-1, 1] \right\} \tag{11}$$

, where $c \in \mathbb{R}^n$ is offset, $\mathbf{G} \in \mathbb{R}^{n \times h}$ is a generator matrix of dependent generators, $G_I \in \mathbb{R}^{n \times q}$ is a generator matrix of independent generators and $E \in \mathbb{N}_{\nu}{}^{p \times h}$ [11].

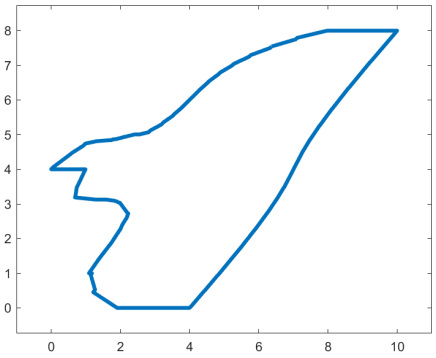


Figure 7: An example of a polynomial zonotope [13]

2.3.3 Level Sets

A level set is the set of all points in the domain where the function takes on a specific constant value. Given a real valued function $\mu(x) : \mathbb{R}^n \rightarrow \mathbb{R}$, the definition for a level set is

$$\mathcal{LS} := \{x \mid \mu(x) \leq 0\} \tag{12}$$

, where any function $\mu(x) \leq c$, where $c \in \mathbb{R}$, can be reformulated to match the above. Level sets are a very general set representation as $\mu(x)$ can be any function. [1].

2.3.4 Star Sets

Intuitively, a star set is a set in which there exists a point from which a line can be drawn to any point in the set without leaving the set [24]. Star sets are flexible as they can define any convex set but also some non-convex sets, such as stars. Additionally, star sets are more efficient under linear maps compared to polytopes.

Given a center point $x_0 \in \mathbb{R}^n$, a set of q vectors $v_1, v_2, \ldots, v_q$ that serve as a basis and a predicate function $P : \mathbb{R}^n \rightarrow \top, \bot$, a generalized star set is formally defined as follows

$$\mathcal{SS} := \left\{ x \mid x = x_0 + \sum_{i=1}^{q} \alpha_i v_i, P(\alpha) = \top \right\} \tag{13}$$

[1].

3 Methodology

This chapter focuses on different set propagation techniques for forward reachability analysis. First, the general framework for set propagation techniques is introduced including over- and under-approximations as well as backward reachability. Finally, cutting-edge set propagation methods for continuous linear and nonlinear systems are introduced.

3.1 Reachability Analysis Using Set Propagation

Dynamical systems are generally analyzed by numerically solving the ordinary or partial differential equations that describe the behavior of a system. Solving these equations requires discretizing the state-space, thus solving them becomes increasingly expensive as the number of state variables increases. An alternative way to tackle the problem is by using set representations such as ellipsoids, zonotopes or star sets to cover all the possible states the system might reach. [1, 2].

Set propagation techniques enable us to compute tight approximations instead of exact numerical solutions, decreasing computational expense. In addition, computing the exact reachable set is often not possible. Over-approximation techniques guarantee that the exact reachable set is included in the approximation. This is useful in applications such as formal verification, where one can check whether the over-approximation intersects with an unsafe set. If not, we can deduce that the unsafe set is not reachable by the system. [1]

Starting with an initial set of states, set propagation algorithms iteratively compute the next possible set of states after an arbitrary small time step. The dynamics of the system and possible input determine the next possible set of states. Disturbances and uncertain parameters in the model can also be considered. The iteration is repeated until the desired time-horizon is reached. The union of computed sets is the reachable set of the system during a selected time horizon. Figure 8 shows an over-approximating set propagation algorithm. [25]

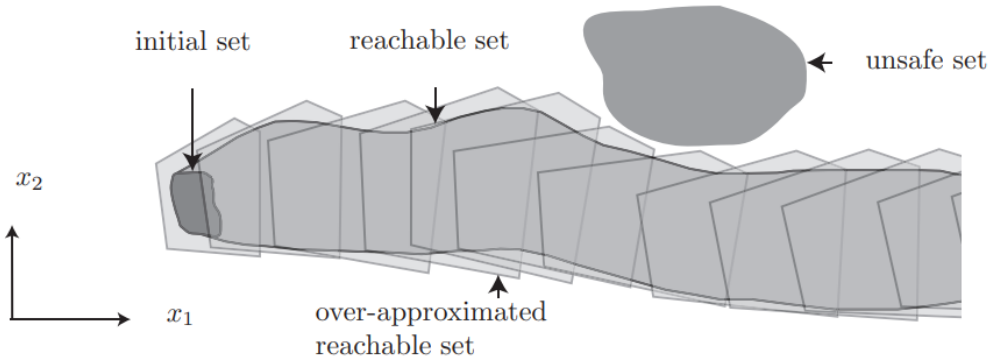


Figure 8: Computation of over-approximated reachable set [10]

3.1.1 Backward Reachability and Inner-approximation

Reachability analysis can be performed backward, starting from a target set. This is similar to forward reachability, but instead of starting with an initial set, we iterate backwards from a target set. The concept of maximal backward reachability involves computing the set of states that can be steered into a certain target set with some feasible control sequence. This is useful in control problems where the target set is a goal set that needs to be reached. To guarantee that the system can reach a goal set, an inner-approximation of the reachable set is required. [14].

Backward reachability can also be used to find the set of states that cannot avoid entering an unsafe set with any possible input. This set is called the minimal backward reachable set. In this case, the backward reachable set is also considered unsafe. This can be applied in formal verification. For instance, in obstacle avoidance, the minimal backward reachable set includes all states from which it is impossible to avoid colliding with the obstacle [14].

Unlike over-approximation methods that compute all possible states a system might reach, inner-approximation methods compute a set of states the system is guaranteed to reach. Inner-approximations are used in control problems where a certain set of states needs to be reached. Inner approximations are not as well researched compared to over-approximations due to their complexity [26].

3.1.2 Applications

Many real-world control problems involve reaching a goal set while avoiding unsafe sets, such as obstacles, and respecting the constraints of the system. These types of problems require formal verification to prove that the system is safe. Reachability analysis is a popular method to formally verify that a given controller satisfies the problem requirements. Over-approximations of the reachable set are used to prove that the system will not enter hazardous sets, whereas inner-approximations can verify that the goal set is reached.

Reachability analysis is used in the estimation of states for safety-critical systems. These set-based observes are more robust than standard methods, such as Kalman filter, as they provide an estimated set that is guaranteed to include the true set of states [1]. Another important application for computing reachable sets is set-based prediction. This is used, for example, to predict the movements of other traffic participants in autonomous vehicles [27].

Computation of backward reachable sets can be applied in controller synthesis [14]. Controller synthesis involves implementing a controller that ensures system safety under realistic operating conditions. In practical applications, factors such as sensor and actuator delays, as well as finite measurement and actuation precision, must be accounted for in the controller design to maintain reliable and safe system behavior [28].

Other applications for reachability analysis are conformance checking [29] and estimation of invariant sets [30]. Conformance checking is used to verify that a mathematical model accurately represents the behavior of the physical system.

Reachable sets can be used to ensure that the model encompasses all possible behaviors of the real system. Estimation of invariant sets is useful in guaranteeing the stability of control. If a state enters an invariant set, it will remain in that set indefinitely. Using reachable sets to estimate invariant set has proven to be more scalable than Lyapunov-based techniques.

3.2 Continuous Linear Systems

Let's consider the dynamics of a linear continuous system:

$$\dot{x}(t) = \mathbf{A}\,x(t) + \mathbf{B}\,u(t), \quad x(t) \in \mathbb{R}^n,\ u(t) \in \mathcal{U} \subseteq \mathbb{R}^m \tag{14}$$

,where $\mathbf{A} \in \mathbb{R}^{n\times n}$ is the system matrix, $\mathbf{B} \in \mathbb{R}^{n\times m}$ input matrix and $\mathcal{U}$ is the set of possible inputs. This input set and the set of initial states are usually assumed convex for linear systems. Consequently, convex set representation such as zonotopes, support functions and ellipsoids can accurately over-approximate the reachable set of linear systems. The basic procedures for reachability analysis using set propagation covered in this section apply for LTI-systems. However, these methods can be extended for systems with uncertain or time-varying parameters.

3.2.1 General Framework of Set Propagation

Most methods for computing reachable sets for continuous linear system rely on time discretization. The time step for the discretization can be made arbitrary small by dividing the time-horizon into N equal parts: $\tau = t_h/(N+1)$. Decreasing the time step size provides more accurate computations. However, the computation time and required memory also increase with respect to N. [1]

Firstly, considering only the system dynamics and initial conditions, homogeneous solution of $\dot{x}(t) = \mathbf{A}\,x(t)$ extended for sets can be computed by a linear map. The first reachable set at time step τ is computed as

$$\mathcal{H}(\tau) = e^{\mathbf{A}\tau}\,\mathcal{X}_0 \tag{15}$$

, where $\mathcal{X}_0$ is the initial set of states and τ is the discretization time. The reachable set during the first time interval is computed by a convex hull of the initial set and Equation 15

$$\mathcal{H}_0 = \mathrm{CH}(\mathcal{X}_0, e^{\mathbf{A}r}\mathcal{X}_0) \oplus \mathcal{F}\mathcal{X}_0 \tag{16}$$

,where $\mathcal{F}\mathcal{X}_0$ is an added enlargement to the set to account for the curvature of trajectories. $\mathcal{F}$ is a correction matrix computed based on the system dynamics. [10]. The computation of the reachable set during first time interval is illustrated below in Figure 9.

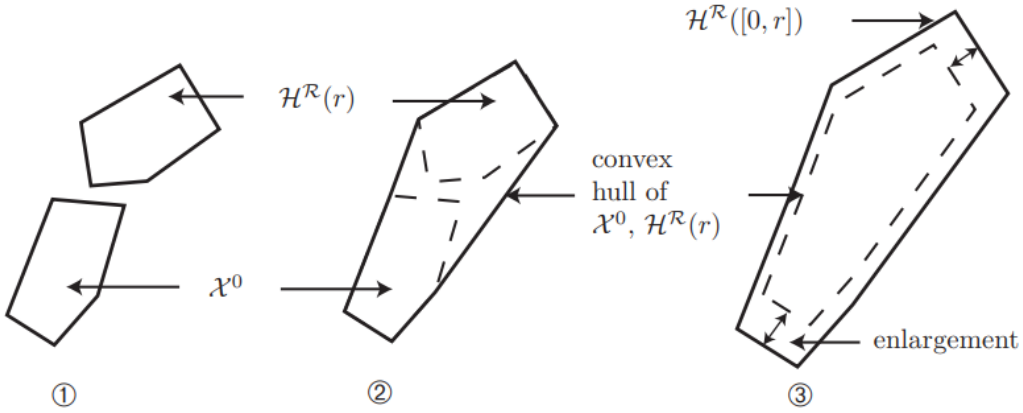


Figure 9: Computation of the reachable set for the first time interval [10]. In the figure, r denotes time step size, $\mathcal{H}^{\mathcal{R}}(\tau)$ is equivalent to $\mathcal{H}(\tau)$ and $\mathcal{H}^{\mathcal{R}}([0,\tau])$ is the reachable set during first time interval.

By exploiting the superposition principle for linear systems, the input to the linear system can be computed separately. Thus, the recurrence relation between reachable sets is

$$\mathcal{H}_{i+1} = e^{\mathbf{A}\tau}\mathcal{H}_i \oplus \mathrm{U} \quad i \in \mathbb{N} \tag{17}$$

,where $\mathcal{H}_i$ is a the set of reachable states during time interval $[i\tau, (i+1)\tau]$, starting from the initial set $\mathcal{X}_0$, $\mathbf{A}$ is the system matrix and U is a set that accounts for input during the time step τ [3]. The whole reachable set during the selected time horizon is obtained by a union of all the reachable sets at each time interval.

3.2.2 Standard Method

The standard approach to computing an over-approximated reachable set for an uncertain linear system involves using Equation 17, to iteratively compute the next reachable set [4]. In practice, the iteration is done using a loop. If the state-space is discretized into N equal parts, the algorithm requires computing the Minskowski sum and linear transformation N times. It can be shown that the time complexity for the reachability algorithm is $O(N^2)$.

The algorithm proposed in [4] uses zonotopes as a set representation, due to the fact that they are closed under Minkowski sum and linear transformations. This makes the algorithm efficient, as these operations are used at every iteration. The order of the zonotopes grows by one with each addition, increasing the computation time. The problem is addressed by reducing the order of the zonotopes by over-approximating them with a lower order zonotope. This method performs well with LTI-systems up to hundred dimensions.

3.2.3 Wrapping Effect- Free Method

Computing reachable sets is done by over-approximating the exact reachable set with a simpler geometric shape, also known as a set representation. When this over-approximated set is used to compute the next reachable set, the approximation errors accumulate during the iteration of Equation 17, possibly causing massive errors with large time horizons [10]. This problem is known as the wrapping effect. Over-approximations can also stem from the fact that the Minkowski sum cannot be computed accurately for certain set representations, such as ellipsoids and zonotopes with bounded order [3]. The wrapping effect is a general term for the propagation of error throughout the iteration.

There are multiple ways to mitigate the wrapping effect in reachability algorithms. In [3], an algorithm that does not suffer from the wrapping effect is introduced. The computation of Minkowski sums increases the complexity of the reachable set at each iteration of Equation 17, and causes over-approximation at each step. Thus, the linear transformations become increasingly expensive to compute and the over-approximation error accumulates. The wrapping effect- free algorithm is similar to the standard method, but it separates the computations required for the next reachable in Equation 17 set as follows:

$$\begin{aligned} X_0 &= \mathcal{H}_0, & X_{i+1} &= e^{\mathbf{A}\tau} X_i, \\ V_0 &= \mathrm{U}, & V_{i+1} &= e^{\mathbf{A}\tau} V_i, \\ S_0 &= \{0\}, & S_{i+1} &= S_i \oplus V_i. \end{aligned}$$

Consequently, the reachable set for the system is $\mathcal{H}_{i+1} = X_{i+1} \oplus S_{i+1}$, where X_{i+1} is the reachable set for the autonomous system without considering input, and S_{i+1} is the reachable set with input. The decomposition of Equation 17 is only possible for LTI-systems, due to the superposition principle [3]. This computation scheme avoids direct iterative applications of Minkowski sums to reduce the computational cost of linear maps and the accumulation of error at each iteration.

3.2.4 Block Decomposition Method

Reachability analysis becomes progressively more costly as the dimension of the system increases. This is mainly due to the increase in computational cost when it comes to performing set operations in higher dimensions. One way to deal with this problem is to break down the large-scale system into smaller subsystems. More specifically, the recurrence relation given in Equation 17 is decomposed into lower-dimensional subsystems, and the reachable set for each subsystem is then computed [31].

With this approach, the set operations used in the algorithm can be performed in lower dimensions, thus reducing the computational complexity. As matrix operations such as matrix exponential can be efficiently computed in high dimension, these operations are computed using the original dimension of the system. This reduces the error caused by the decomposition. After computing the reachable sets in low-dimensional subsystems, the sets are composed to form the reachable set of the original system.

The method introduced in [31], is capable of computing the reachable set for systems with more than ten thousand state variables in reasonable time. The method proves to be up to a hundred times faster than state-of-the-art techniques that do not use decomposition. The speedup comes at the cost of accuracy, but according to the paper the over-approximation is no larger than 22% in all but one of the case studies.

3.2.5 Krylov Subspace Method

Computing $e^{\mathbf{A}\tau}\mathcal{H}_i$ for high-dimensional systems proves to be computationally expensive. One of the most successful solutions to this problem has proven to be an order reduction technique called Krylov methods [8]. These methods employ the Krylov subspace to efficiently approximate $e^{\mathbf{A}\tau}\mathcal{H}_i$. The simplest and most popular approach to approximating the matrix exponential is called the Arnaldi algorithm, which is used in the aforementioned research paper. The algorithm is based on the wrapping effect-free method, but it approximates the computation of $e^{\mathbf{A}\tau}\mathcal{H}_i$ using the Krylov subspace.

In real-world systems, inputs cannot be measured accurately due to disturbances, such as sensor noise. In [8], a method for computing reachable sets for high-dimensional systems with bounded arbitrarily varying input is introduced. This method can handle systems with arbitrarily varying inputs with thousands of state variables. The Krylov method proves to be up to hundred times faster compared to standard methods [8].

3.3 Nonlinear Systems

This section covers set propagation methods for nonlinear systems with dynamics defines as

$$\dot{x}(t) = f(x(t), u(t)), \quad x(t) \in \mathbb{R}^n,\ u(t) \in \mathbb{R}^m \tag{18}$$

, where $f : \mathbb{R}^n \to \mathbb{R}^n$ is a nonlinear function, $x(t)$ is the state vector and $u(t)$ is the input vector [32]. Both of the methods in this section resort to abstraction techniques to compute the reachable for nonlinear systems. This is because computing reachable sets for nonlinear systems is challenging, as useful properties, such as linear mapping and the superposition principle no longer apply [27].

3.3.1 Conservative Linearization Method

One way to compute reachable sets for a nonlinear system, is by locally linearizing the system and then using the same methods as for linear systems. In [33], the system is divided into equal segments, and linearization is performed within each segment. A more adaptive approach partitions the state space on the fly, based on the the geometry of the reachable set [34].

Linearization causes additional error in the computation, which is considered as an additional enlargement of the local reachable set. To control this error, the reachable sets are split until the linearization error is within admissible bounds. This technique guarantees that the reachable set of the original nonlinear system is contained by the reachable set of the linearized system. The computation scheme for during one time

interval is illustrated in Figure 10. The method proposed in [34], performs well for nonlinear systems with up to thirty dimensions.

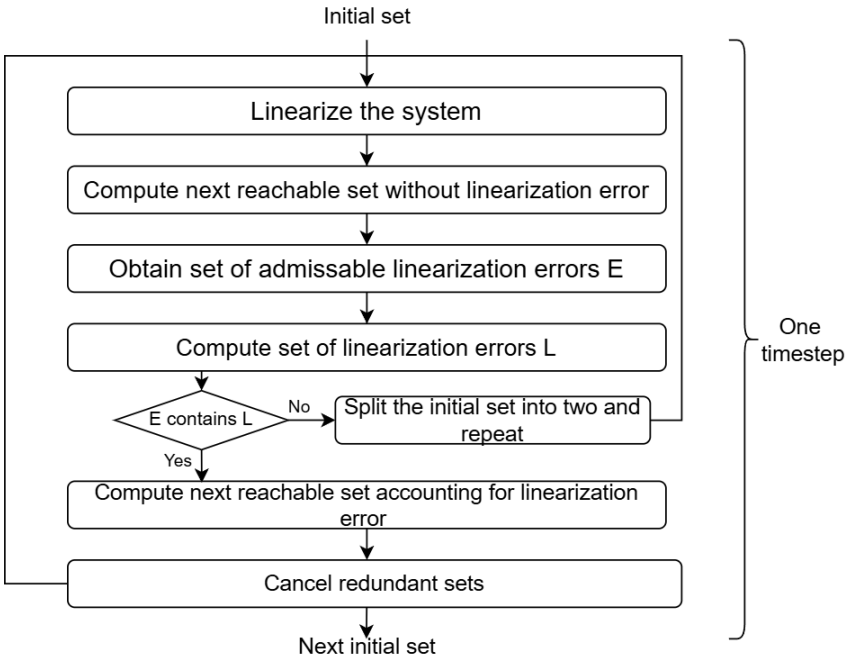


Figure 10: Computation scheme of the reachable set for one time interval using the linearization method. Simplified figure from [34]

3.3.2 Conservative Polynomialization Method

If the system is highly nonlinear, the linearization method becomes costly as the reachable sets have to be split many times until the linearization error is small enough. Consequently, the number of linearizations required grows exponentially.The conservative polynomialization method [11] improves the previous method by also abstracting the nonlinear dynamics of the system by a Taylor expansion. As a result, splitting of the sets can be avoided more often.

As the reachable sets for nonlinear systems are generally nonconvex, a set representation called polynomial zonotopes is proposed in [11] to accurately represent the reachable sets. This further reduces the need for splitting the sets, as the error becomes smaller.

A further improvement of the former method is presented in [15] through the introduction nonconvex sparse polynomial zonotopes. The use of this set representation effectively mitigates the wrapping effect during reach set computations for nonlinear systems. Moreover, sparse polynomial zonotopes are closed under useful set operations, and these operations have no more than polynomial complexity.

3.3.3 Adaptive Parameter Tuning

Set propagation algorithms have multiple crucial parameters that significantly affect the resulting reachable set. These parameters include time step size, order of the Taylor series and order of the set representations. For example, time step size has a huge impact on the accuracy of the iteration. By reducing the time step size, the error of the algorithm can be made arbitrarily small. The first adaptive parameter tuning algorithm is proposed in [32]. The algorithm automatically tunes each parameter separately, providing a modular approach that can be adapted for different set representation and approaches.

The results demonstrate that adaptive parameter tuning provides fast and accurate approximations of reachable sets when using linearization or polynomialization approaches. Furthermore, the tightness of the approximations is comparable to that achieved by algorithms tuned by experts. The key advantage of adaptive parameter tuning is that it does not require a human to make adjustments to the parameters. Consequently, this method saves time since manual tuning typically requires multiple runs. Table 1 summarizes all set propagation algorithms introduced in this section.

Table 1: A table comparing set propagation algorithms

Algorithm	**Description**	**System type**	**Remark**
Standard approach [4]	Computes reachable set by iterating equation (16)	Linear	Uses zonotopes as a set representation
Wrapping Effect - Free [3]	Restructures the computations to avoid overapproximations	Linear	Free of the wrapping effect
Block Decomposition [31]	Decomposes state space into subsystems	Linear	Set operations performed in lower dimension
Krylov Subspace [8]	Uses Krylov subspace for computations	Linear	Handles arbitrarily varying inputs
Linearization [34]	Linearizes nonlinear dynamics locally	Nonlinear	Linearization error controlled by splitting reachable sets
Polynomialization [11]	Nonlinearities abstracted with Taylor approximation	Nonlinear	Uses a non-convex set representation

4 Case Studies

This section compares the performance of the set propagation methods introduced in the previous section. The advantages and limitations of these methods are studied by testing them with different benchmark systems that resemble real-world applications. The computations for linear systems were carried out using MATLAB-based reachability toolbox called CORA [13] on a system equipped with Apple 3.2 GHz 8GB memory M1 processor. For nonlinear systems, the computations were performed using a computer with an Intel 10-core i5 processor running at 2.5GHz and 16GB of RAM.

4.1 Performance Analysis

The performance of different set propagation methods is analyzed by comparing the computation time and the tightness of the approximation. The tightness of the computed reachable sets is calculated by comparing them with simulations of the systems. Simulations are used as comparison as it is not possible to compute the exact reachable set for any of the systems tested. The metric used to measure the tightness is first proposed in [26]. The main idea is that for an n-dimensional system, the reachable set and the simulations are approximated with an n-dimensional box. After that, the width of the boxes is compared in every direction, and the smallest width ratio is selected. The equation for this tightness measure is

$$\gamma_{\min} = \min_{i \in \{1,\dots,n\}} \frac{d_i(\,\mathbf{box}(\,\mathbb{R}_{sim}(t_f))}{d_i(\,\mathbf{box}(\,\mathbb{R}(t_f))} \tag{19}$$

, where $\mathbf{box}(\,\mathbb{R}_{sim}(t_f))$ and $\mathbf{box}(\,\mathbb{R}_{sim}(t_f))$ are the box approximations of the simulations and the reachable set, respectively. The box approximation encloses the given set by computing upper and lower bounds of the set. t_f (final time) is the time at which the approximation is computed. d_i is essentially the extent of the box approximation in a certain direction. The closer the value of $\gamma_{\min}$ is to 1, the more accurate the approximation becomes.

4.2 RLC-circuit

Let us first consider a simple RLC-circuit example from [35]. The circuit is comprised of a resistor, coil and capacitor. The resistance, inductance and capacitance of the components are

$$R = 2.5\Omega, \quad L = 3\text{mH}, \quad C = 2\text{mF}$$

, respectively. The system is linear and its system matrix and input matrix are defined as

$$\mathbf{A} = \begin{bmatrix} -\frac{1}{RC} & \frac{1}{C} \\ -\frac{1}{L} & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix}$$

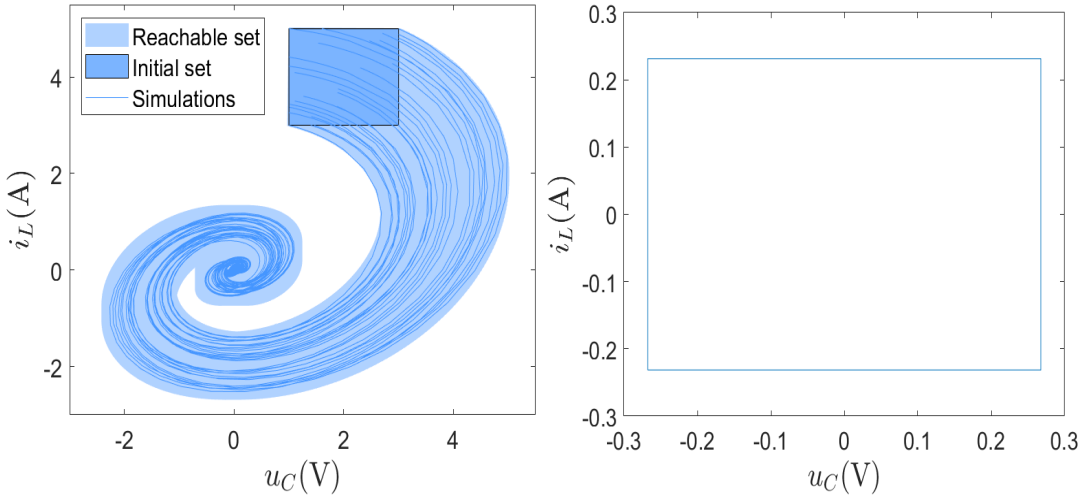
Thus, the state-space representation for the circuit is

$$\begin{bmatrix} \dot{u}_C(t) \\ \dot{i}_L(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{RC} & \frac{1}{C} \\ -\frac{1}{L} & 0 \end{bmatrix} \begin{bmatrix} u_C(t) \\ i_L(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} u_I(t)$$

, where $u_C(t)$ is voltage at the capacitor, $i_L(t)$ is the current at the coil and $u_I(t)$ is the input voltage with values ranging between [-0.1,0.1]V. Table 2 shows the runtime and tightness of each method for the RLC-system. Figure 11 displays the reachable set for the RLC-system computed using the Wrapping-Free method.

Table 2: Computation times for the RLC-model in seconds

Algorithm	**Runtime (s)**	γ_{min}
Standard [4]	6.1	0.17
Wrapping-Free [3]	8.9	0.17
Decomposition [31]	22.1	0.17
Krylov [8]	24.0	0.17



(a) Reachable set for the whole time horizon **(b)** Final reachable set after 2 seconds

Figure 11: The whole reachable set for 2 second time horizon (a) and final reachable set after 2 seconds (b) for the RLC-system

We observe that both states $u_C(t)$ and $i_L(t)$ converge to zero overtime. This is because the input voltage has a mean value of zero and the circuit does not have a DC component to sustain voltage.

4.3 Building Model

The building model represents the Los Angeles University Hospital as a dynamical system [36]. Specifically, it models how the structure of the building responds to dynamic forces, such as earthquakes. The building has 48 state variables, corresponding to the 3 degrees of freedom for the 8 floors. The degrees of freedom are displacement in the x-y plane and rotation. Both the positions and their rates of change are included in the state representation, resulting in a total of 48 state variables. Table 3 displays results for the algorithms tested on the model.

Table 3: Computation times for the Building model in seconds

Algorithm	**Comp Time (s)**	$\gamma_{\min}$
Standard [4]	1.7	0.71
Wrapping-Free [3]	2.9	0.67
Decomposition [31]	1.5	0.31
Krylov [8]	9.7	0.48

4.4 International Space Station Model

The International Space Station (ISS) model describes the structural dynamics of a service module and how it responds to inputs [37]. It is a continuous LTI-system that has 270 state variables and 3 inputs. This means that the system matrix of the model is $A^{270\times270}$ and the input matrix is $B^{270\times3}$. Table 4 presents the results for the ISS-model.

Table 4: Computation times for the ISS-model in seconds

Algorithm	**Runtime (s)**	$\gamma_{\min}$
Standard [4]	15.3	0.85
Wrapping-Free [3]	13.1	0.95
Decomposition [31]	3.68	0.79
Krylov [8]	4.61	0.78

4.5 Heat3D Model

The Heat3D model models the heat transfer of a block in three dimensions [37]. The dynamics of the system are therefore given by the PDE called the heat equation. The 3d block is discretized into a $k \times k \times k$ grid, resulting in a linear system. The system models the change in temperature at each point on the grid. The dimension of the system is k^3, and it can be increased by dividing the block into more parts. Thus, the model is an excellent tool to test the scalability of different methods to higher-dimensional systems. Table 5 presents the results for a 125-dimensional system, which is obtained by discretizing the block into a $5 \times 5 \times 5$ grid. In Figure 12, the scalability of different methods is illustrated using the Heat3D model.

Table 5: Computation times for the 125-dimensional 3D heat model in seconds

Algorithm	Runtime (s)	$\gamma_{\min}$
Standard [4]	15.7	0.27
Wrapping-Free[3]	14.7	0.29
Decomposition [31]	9.5	0.27
Krylov [8]	7.4	0.27

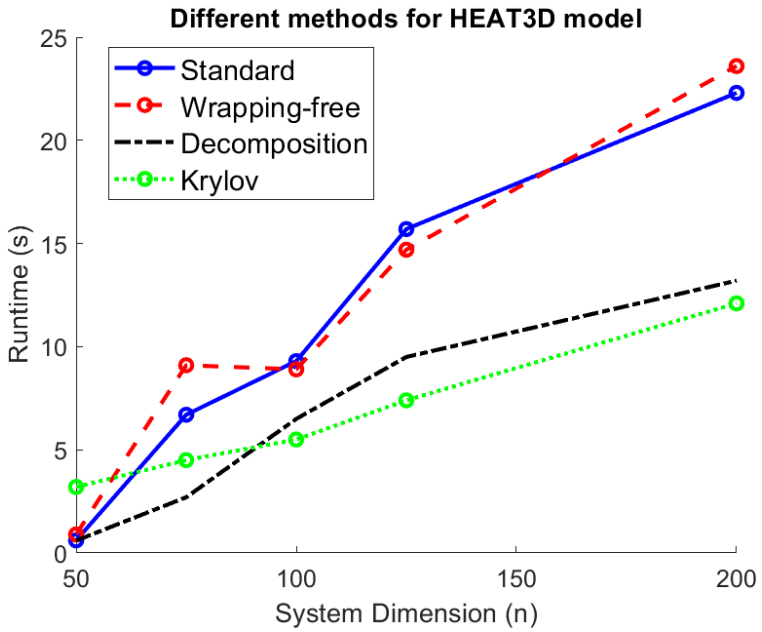


Figure 12: Runtime of Heat3D model for different methods with increasing system dimension

4.6 Van der Pol oscillator

The Van der Pol oscillator is a nonlinear damping oscillator system. It was originally used to model triode circuits, but it has since been used in various applications, such as modeling heartbeats [38], limit cycles in control systems [39] and robotics [40]. The Van der Pol oscillator dynamics are defined by following equations

$$\dot{x_1} = x_2 \tag{20}$$

$$\dot{x_2} = \mu(1 - x_1^2)x_2 - x_1 \tag{21}$$

,where μ is a scalar parameter that controls the nonlinearity and the damping of the system. In the tested model, μ is equal to one. Table 6 summarizes the numerical results of the Linearization and Polynomialization methods tested on the model. Figure 13 illustrates the reachable sets computed by the methods.

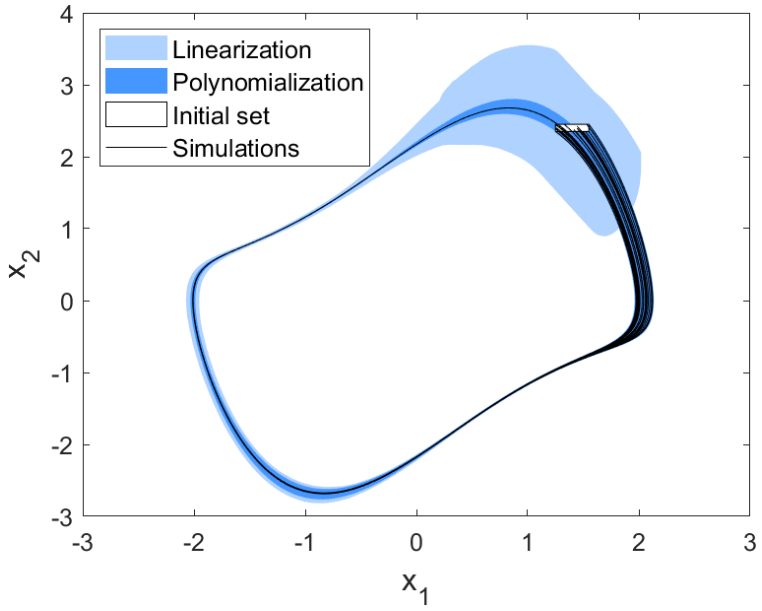


Figure 13: Reachable sets of Van der Pol oscillator computed using different methods

Table 6: Computations for Van der Pol oscillator

Algorithm	**Runtime (s)**	$\gamma_{\min}$
Linearization [34]	6.21	0.14
Polynomialization [11]	39.73	0.52

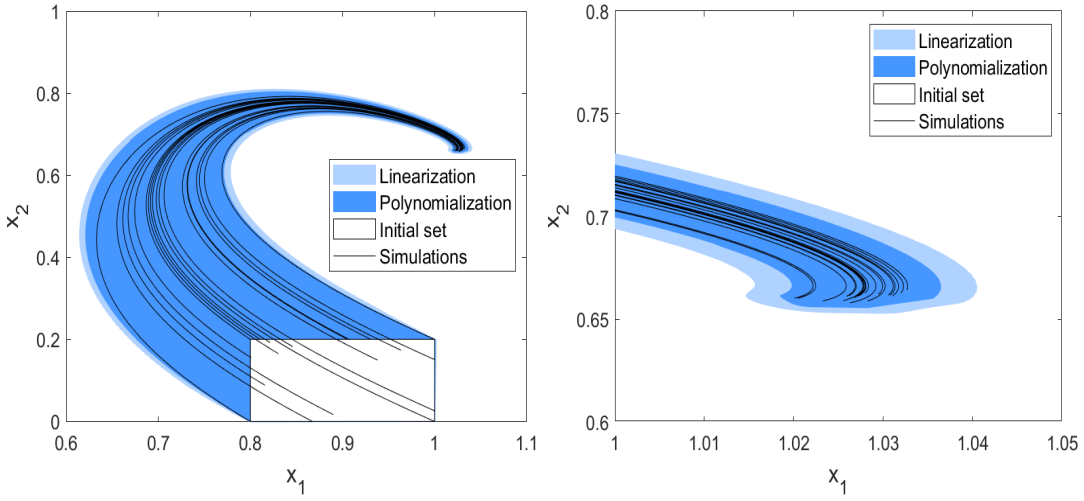
4.7 Brusselator

Brusselator is a model that describes an oscillating autocatalytic chemical reaction [41]. The model has a variety of applications such as pattern formation in biology, non-equilibrium thermodynamics and control theory. It can be extended to explain how spatial structures form in animal coats, seashells, and tissues [42]. The Brusselator model can also be used to study the control of chaotic systems [43] and synchronization of oscillators [44]. Dynamics of the Brusselator can be defined as

$$\dot{x_1} = 1 - (b+1)x_1 + ax_1^2x_2 \tag{22}$$

$$\dot{x_2} = bx_1 - ax_1^2x_2 \tag{23}$$

,where a and b are real-valued positive constants, in this case a=1 and b=1.5, while x and y denote the concentrations of the reactants [41]. Table 7 shows the runtime and tightness of the tested methods for the Brusselator model. Figure 14 displays the reachable sets for the Brusselator obtained by using Linearization and Polynomialization methods.



(a) The reachable set of the Brusselator for the whole time horizon

(b) The final reachable of the Brusselator after 6 seconds

Figure 14: The whole reachable set for 6 second time horizon (a) and final reachable set after 6 seconds (b) for the Brusselator

Table 7: Computations for Brusselator

Algorithm	**Runtime (s)**	$\gamma_{\min}$
Linearization [34]	2.76	0.42
Polynomialization [11]	14.52	0.63

4.8 Rössler Attractror

The Rössler attractror is a system that is defined by three nonlinear differential equations. It behaves periodically with certain parameters, and chaotically for others. The Rössler attractor has been used to study chaos control methods [45] and chaos-based encryption [46]. The standard form of the Rössler system is given by equations:

$$\dot{x_1} = -x_2 - x_3 \tag{24}$$

$$\dot{x_2} = x_1 + ax_2 \tag{25}$$

$$\dot{x_3} = b + x_3(x_1 - c) \tag{26}$$

, where x, y and z are the state variables of the system and a,b and c constant parameters. The model is tested with parameters a = 0.2, b = 0.2 and c = 5.7. These parameters cause the model do exhibit chaotic behavior. Table 8 provides the results of the tests.

Table 8: Computations for Rössler

Algorithm	**Runtime (s)**	$\gamma_{\min}$
Linearization [34]	8.05	0.19
Polynomialization [11]	27.49	0.61

4.9 9-dimensional Genetic Oscillator

To test the scalability of the nonlinear approaches, a 9-dimensional genetic oscillator model is considered [47]. The model consists of nine differential equations describing the fundamental principles of circadian clocks. Circadian clocks are used by many organisms to keep track of daily time. Table 9 shows the runtime and tightness of tested algorithms. Below are the differential equations that govern the system.

$$
\begin{aligned}
\dot{x}_0 &= 50x_2 - 0.1x_5 \\
\dot{x}_1 &= 100x_3 - x_1x_1 \\
\dot{x}_2 &= 0.1x_5 - 50x_2 \\
\dot{x}_3 &= x_1x_5 - 100x_3 \\
\dot{x}_4 &= 5x_2 + 0.5x_0 - 10x_4 \\
\dot{x}_5 &= 50x_4 + 50x_2 + 100x_3 - x_5(0.1x_0 + x_1 + 2x_7 + 1) \\
\dot{x}_6 &= 50x_3 + 0.01x_1 - 0.5x_6 \\
\dot{x}_7 &= 0.5x_6 - 2x_5x_7 + x_8 - 0.2x_7 \\
\dot{x}_8 &= 2x_5x_7 - x_8
\end{aligned}
$$

Table 9: Computations for 9D-oscillator

Algorithm	**Runtime (s)**	$\gamma_{\min}$
Linearization [34]	6.06	0.06
Polynomialization [11]	47.07	0.34

4.10 Discussion

Tables 10 and Table 11 summarize the results obtained for the case studies of LTI-systems and nonlinear systems, respectively. The main observations for LTI-systems are that the decomposition and Krylov methods have notably faster runtime with higher-dimensional system while the accuracy γ_{min} of these methods is lower compared to the Standard and the Wrapping-free methods. For nonlinear systems, the Polynomialization approach yields more accurate results compared to the Linealization, but its runtime is considerably higher.

For LTI-systems, the parameters for the algorithm have to be tuned manually. The algorithm parameters, such as time step size, time horizon, and the number of Taylor terms used for the approximation of the matrix exponential $e^{\mathbf{A}\tau}$, have a significant effect on the computation time and the accuracy of the computed reachable set. The decomposition method has its own specific parameter that defines how the system is partitioned. Similarly, the Krylov method has its own parameters, which affect the results. Furthermore, the structure of every system is different, thus the results for different systems are not totally comparable. To summarize, achieving optimal results with the algorithms for linear systems requires careful tuning of the parameters and deep knowledge of the system.

For nonlinear systems, all algorithm parameters can be tuned adaptively with the adaptive parameter tuning approach. Consequently, the results for nonlinear systems correspond with results in the literature [32]. However, the size and shape of the initial set have a massive impact on the computation time and tightness of the reachable set. This is especially evident when dealing with higher-dimensional systems. Algorithms used for nonlinear systems can also be tuned manually. For instance, the scalability of the algorithms can be improved by decreasing the number of Taylor terms. This makes the algorithm less accurate but computationally less expensive.

Table 10: A table comparing set propagation algorithms for linear systems: n denotes system dimension, t_f is the time horizon in seconds, δ is time step size, time indicates runtime in seconds and γ_{min} tightness of the final reachable set

System	n	t_f	δ	Standard		Wrapping		Decomp		Krylov	
				Time	γ_{min}	Time	γ_{min}	Time	γ_{min}	Time	γ_{min}
RLC	2	2	10^{-5}	6.1	0.17	8.9	0.17	22.1	0.17	24.0	0.17
Building	48	20	10^{-3}	1.7	0.71	2.9	0.67	1.5	0.31	9.7	0.48
ISS	270	20	10^{-3}	15.3	0.85	13.1	0.95	3.68	0.79	4.61	0.78
Heat3d	125	40	10^{-3}	15.7	0.27	14.7	0.29	9.5	0.27	7.4	0.27

Table 11: Comparison of the Linearization and Polynomialization approaches using the adaptive parameter tuning algorithm: n denotes the system dimension, t_f is the time horizon, X_0 is the initial set, time indicates runtime in seconds and γ_{min} tightness of the final reachable set

System	n	t_f	X_0	Linearization		Polynomialization	
				Time	γ_{min}	Time	γ_{min}
Van Der Pol	2	6.74	$[1.25, 1.55]$ $[2.35, 2.45]$	6.21	0.14	39.73	0.52
Brusselator	2	6	$[0.8, 1.0]$ $[0.0, 0.2]$	2.76	0.42	14.52	0.63
Rössler	3	5	$[-0.20, 0.20]$ $[-8.60, -8.20]$ $[-0.20, 0.20]$	8.32	0.43	31.45	0.70
9D	9	1	$[0.99, 1.01]^n$	6.06	0.06	47.07	0.34

5 Conclusions

The comparison of the chosen set propagation methods proved that certain methods perform better for different type of systems. Namely, the decomposition and Krylov methods scale better for large-scale systems, albeit at the expense of reduced accuracy. The standard and wrapping-free methods provide more accurate results. However, their computational cost grows rapidly with the system dimension. For nonlinear systems, the algorithm parameters can be tuned adaptively with the adaptive parameter tuning approach. The linearization approach proved to be faster but less accurate, whereas the polynomialization approach results in a significantly tighter reach set. However, the polynomialization approach is several times slower. For real-life applications, the choice of the set propagation method depends on the needs of the application. The factors to consider are the computation time and the accuracy of the computed reachable set.

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